

Name: Dylan M

Date: 3/17/2026

Class: Engineer Design

MyDesign Portfolio

Element C

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

Our design MUST block airflow through the dust port system.

If we are not able to block airflow through an open port, the extra air flow will decrease the efficiency of the A.C.T dust collecting system.

Our stakeholder (**Agrees** or **disagrees**) because preventing airflow is key to letting the dust collector run most effectively when the p[orts are closed and not in use.

Design Requirement #2 (HIGH Priority)

Our design must cover the entire 3 inch open port to ensure air flow is blocked .

We must design our idea to cover the entire 3inch diameter of the port so there will be no airflow that can disrupt the flow and efficiency of the dust collector.

Our stakeholder (**Agrees** or **disagrees**) because if the idea is not seal tight, air will flow through the gasket and create less efficiently in our design.

Design Requirement #3 (HIGH Priority)

Our design must be able to open or close, or be removable so that the dust port can either be open or closed

If our design fails to be removable, or open and closed, the ports will not be able to be used, and we must use these ports for woodshop and other machinery..

Our stakeholder (**Agrees** or **disagrees**) in case a word piece gets stuck, or the ports need to be removed and cleaned, our idea should be removed for safety and cleaning

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Design Requirement #4 (HIGH Priority)

Our design must be affordable, such as pricing under \$5 a piece to ensure we are efficient with our budget.

We need to be able to create a design that is affordable, so the school can afford to buy several of them and buy more whenever replacements are needed

Our stakeholder (**Agrees** or **disagrees**) because this will make it easier for the school to buy equipment for the woodshop

Design Requirement #5 (HIGH Priority)

Our design must ensure all dust is being sucked into the open port, and not block any dust from getting into the system.

This ties into our prior research because all dust needs to be sucked into the port to create the cleanest environment possible in and around the woodshop at the school.

Our stakeholder (**Agrees** or **disagrees**) because the ports having max efficiency will maximize the cleanliness of the woodshop

Design Requirement #6 (LOW Priority)

Our design must look visually appealing to our stakeholders

If our design not only works great but looks amazing, It will create a better looking room for students to enjoy during their woodshop class

Our stakeholder (**Agrees** or **disagrees**) because the ports must look appealing and clean as it will create a better and safe environment for all.

Design Requirement #7 (LOW Priority)

Our design must have a space to put your fingers or have easy access to open and close or remove the design from the port incase the port must be cleaned or taken out of the system

This idea will allow easy access to open and close the ports for students to use these ports in woodshop

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Our stakeholder (**Agrees** or **disagrees**) because it allows the ports to be used by students and not just teachers having to open and close.